FIGURE 1 · METHODOLOGY

The SHARP–AEGIS pipeline

SHARP measures the smallest honest model; AEGIS governs whether it is fit for its job; a human signs every gate. The machine only recommends.

1 DATA INTAKE & SCOPING

"What is the target, how many defaults, any protected attributes?"

Size by the scarcest resource: EPV = defaults ÷ features. Firm attrs → fairness N/A; person attrs → fairness live.

2 SHARP — THE FEATURE FUNNEL

Each stage removes a different kind of lie. Pre-registered, fixed across all datasets.

1. **Screen** — drop obvious noise and leakage
2. **Correlation grouping** — collapse look-alike features into one "player"
3. **Shapley ranking** — credit each player fairly over every team it could join
4. **Shadow-null** — keep only features that beat an injected random decoy
5. **Parsimony** — smallest set with DeLong p ≥ 0.05 and rel-Gini loss ≤ 5%

◆ **columns lock → THE model is fit here (WOE-logistic on the final columns)**

3 ALWAYS-ON SCORING HYGIENE

Isotonic calibration
"we said 10%" ≈ 10-in-100 default

PD floor 0.03%
no zero-risk borrower (Basel CRE 32.3)

4 AEGIS — THE GATE LADDER

Machine recommends + evidence → human signs → hash-chained ledger. Gates decide governance, not analytics.

G0	**Eligibility & scope**	what may we model? protected attrs & materiality tier
G1	**Data sufficiency**	is there enough — EPV — to proceed honestly?
G2	**Feature governance**	features legitimate? no proxies, no leakage
G3	**Model & method**	confirm the pre-registered method — do not retune
G4	**Performance & calibration backtest**	per-grade Wilson CIs, Hosmer–Lemeshow, ECE (held-out 1/5)
G5	**Deliverables**	decisioning note + 6-case pack; no cutoff/pricing invented
GF	**Final verdict**	are the weaknesses tolerable at this model's tier?

sealed by a hash chain (genesis → G_n → …) so the record cannot be edited after signing

continued →

FIGURE 1 · CONTINUED

The governed verdict — and the eleven runs

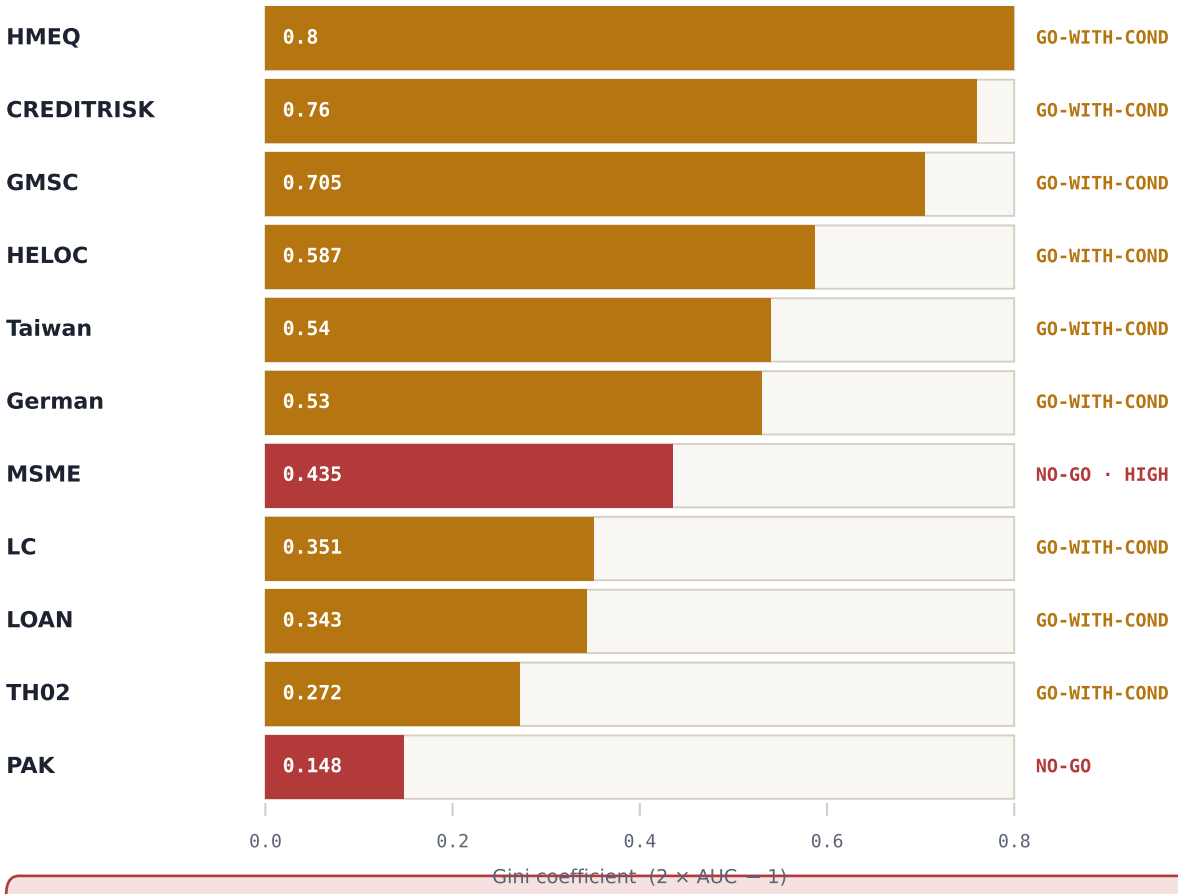
5 THE GOVERNED VERDICT

Read off fitness-for-tier — strictness scales with how much the model matters (LOW → MEDIUM → HIGH).

GO	GO-WITH-CONDITIONS	NO-GO
fit for its tier as-is	usable with named monitoring / limits	refused — by judgment or by rule

6 ELEVEN RUNS — AND WHY GINI DOES NOT DECIDE

Each model's Gini (rank-ordering power) vs its verdict. No single horizontal line separates the colours.



The crossovers

MSME (Gini 0.435) is a NO-GO, yet LC (0.351), LOAN (0.343) and TH02 (0.272) — all weaker — are GO-WITH-CONDITIONS. MSME failed on EPV ≈ 1.2 at HIGH tier; LC has the highest EPV (~245) yet only modest Gini. The verdict tracks EPV, data quality, fairness and tier — not Gini, and never a single accuracy cutoff.

Figure 1. The SHARP–AEGIS governance pipeline. SHARP selects the smallest defensible feature set and reports discrimination with a confidence interval; AEGIS governs deployment through a human-signed, hash-chained gate ladder. Across eleven credit datasets the same process produced nine GO-WITH-CONDITIONS and two NO-GO verdicts — one by human judgment (PAK: redlining proxies), one by rule (MSME: HIGH-tier data-sufficiency bars) — demonstrating risk-proportionate rigor, not one cutoff.

Gini values are per-run point estimates from the recorded runs; grade-band edges require an external/stated internal scale citation before publication.

Appendix A — Reproducibility reference

Every statistical and machine-learning method, threshold and parameter used in the SHARP –AEGIS pipeline is recorded here, so the eleven results on pages 1 –2 can be reproduced from this document alone. All values are FIXED across every dataset unless stated otherwise. Authoritative sources: METHODOLOGY.md (SHARP v3, the pre-registered spec) and AEGIS_GOVERNANCE.md. SHARP measures; AEGIS governs; a human signs each gate.

A.0 Learner, value function & global parameters

Learner (default)	WOE-logistic — penalised logistic regression on monotonic weight-of-evidence inputs; deterministic, hyperparameter-insensitive. Secondary/robustness: XGBoost (report both + Kendall's τ).
Value function v(S)	cross-validated AUC on exactly the columns in S; v(∅) = 0.5 by convention. Alternatives: partial_auc, brier_skill.
Cross-validation	5-fold StratifiedKFold, shuffle=True (GMSC is sorted by target — shuffling is mandatory).
Seed (every stage)	random_state = 42.
Class weighting	scale_pos_weight OFF by default (leaving it on inflated mean PD 4.3× on GMSC).
Primary metric	Gini = 2·AUC − 1 (AUC reported alongside for cross-literature comparability).

A.1 One-call reproduction

```
import sys; sys.path.insert(0, 'ml_models')
import pandas as pd, v3_pipeline as v3
out = v3.run_pipeline(pd.read_csv('<dataset>.csv'), target='target', dataset_name='<NAME>')
print(out['report'])   # Stage 0-4, selected set, metrics
# deployed scorer (reason codes + PD floor):
import missing_governance as mg
scorer, report = mg.build_scorer(df, 'target', selected_cols,
        mandatory_columns=(...), incomplete_policy='refer',
        random_state=42, pd_floor=0.0003, pd_cap=0.999)
proba = scorer.score(new_df)['probability']
```

A.2 Gate → test crosswalk — which test runs at which AEGIS gate

Each AEGIS gate G0–GF REVIEWS and signs off one part of the SHARP pipeline. This table is the index: given a gate, the exact tests performed there; given a test, the gate that owns it. Full detail for each follows in A.3–A.9. Gates G5 and GF add no new statistical test — they govern deliverables and the final judgment.

G0 Eligibility & scope (↔ Stage 0)	eligibility tagging (prohibited / proxy_risk); availability tagging; EPV = events ÷ features; sentinel / leakage audit (event-rate lift ≥ 3×); materiality tiering. No performance test yet.

G1 Data sufficiency (↔ Stage 1)	EPV ≥ 10 (Peduzzi); Information Value screen IV ≥ 0.02 (Siddiqi) + IV bands; Cramér's V (descriptive); monotonic WOE binning; bin-count sensitivity; χ^2 min expected cell ≥ 5 (Cochran); Mann-Whitney U + bootstrap-CI model check on escalation.
G2 Feature governance (↔ Stage 2)	SPVIM Shapley importance; correlation grouping (Spearman $\lvert\rho\rvert \geq 0.7$, VIF > 10); shadow-null retention (Shapley > 95th pct of shadows AND confirmation ≥ 0.80); Monte-Carlo SE. Benjamini-Hochberg p / Cohen's d are DIAGNOSTICS ONLY.
G3 Model & method (↔ Stage 3)	parsimony curve; DeLong paired-AUC test (p ≥ 0.05 = indistinguishable from full); relative Gini loss ≤ 5%; bootstrap 95% CI on Gini; KS, pAUC, Brier reported per k.
G4 Performance & calibration (↔ Stage 4 + backtest)	80/20 stratified backtest; per-grade observed vs predicted with Wilson 95% CI; Hosmer-Lemeshow GoF (p < 0.05 rejects); ECE; isotonic calibration + reliability diagram; fairness AIR / EEOC four-fifths ≥ 0.80; ECOA/Reg B EDDSS age check.
G5 Deliverables / deployment	no new statistical test: exact additive WOE reason codes; PD floor 0.03% / cap 0.999; mandatory-field gate; 6-case acceptance pack; decisioning note (no cutoff or pricing invented).
GF Final verdict	no new test: MAC Go/No-Go against tier-acceptance bars (HIGH tier e.g. EPV ≥ 10, AUC ≥ 0.80, HL pass, no thin grades); weakness-tolerability judgment; hash-chain seal.

A.3 Stage 0 — Feature eligibility & data governance · gate G0 (before any statistics)

Eligibility tags	permissible / prohibited (SEX, AGE or YOB, MARRIAGE, spouse-income SINC) / proxy_risk (EDUCATION, geography, dependents NKID, housing or RES) / unclear. Prohibited & proxy are RETAINED for literature comparability with mandatory warning + disclosure — never an endorsement of live use.
Availability tags	origination / behavioural / near_target / external_score. near-target and external-score features are reported WITH and WITHOUT them.
Sentinel / leakage audit	non-measurement codes recoded as explicit indicators; any sentinel group with event-rate lift ≥ 3× flagged as a leakage candidate.
Missing-value policy	declared up front: missing_policy ∈ {as_learned, adverse_floor} with floor_columns; mandatory_columns with incomplete_policy ∈ {reject, refer, adverse_score}.

A.4 Stage 1 — Univariate screening · reviewed at gate G1

Primary screen (decides)	Information Value IV ≥ 0.02 (Siddiqi “weak or better”) — the single deciding criterion.

IV bands	< 0.02 useless · 0.02–0.1 weak · 0.1–0.3 medium · > 0.3 strong.
Cramér's V (descriptive)	< 0.1 negligible · 0.1–0.3 weak · 0.3–0.5 moderate · > 0.5 strong. Reported but DECIDES NOTHING.
Binning	monotonic WOE, minimum bin = 5% of rows; discrete/count columns use raw levels (rare pooled).
Non-monotonic re-route	if removing the monotonicity constraint recovers ≥ 20% more IV → route to single-column model check, not decided on IV.
Bin-count sensitivity	IV recomputed at WOE and 5/10/20 quantile bins; if the IV ≥0.02 verdict flips → binning_sensitive → model check.
Reliability gate	χ^2 minimum expected cell count ≥ 5 (Cochran) before the binned statistic is trusted.
Escalation on gate fail	orderable → Mann-Whitney U; nominal → single-column model check.
Model check verdict	Gini with bootstrap 95% CI; CI must exclude zero to confirm "real".

A.5 Stage 2 — Order-invariant importance (SPVIM + shadow-null) · reviewed at gate G2

Method	Shapley Population Variable Importance Measure (SPVIM; Williamson & Feng 2020) — refit per subset, value = CV predictiveness. Not SAGE.
Correlation grouping	hierarchical clustering on distance 1 − |Spearman ρ|, cut at |ρ| = 0.7; each cluster = ONE Shapley player. VIF > 10 flagged. Sensitivity at ρ ∈ {0.6, 0.7, 0.8}.
Shadow-null retention	add ≥ 10 shadow players (row-permuted copies — Boruta device). Confirm a real player iff Shapley > 95th percentile of the shadow distribution AND bootstrap confirmation frequency ≥ 0.80. No p-value; no multiple-testing correction.
Exact vs sampled	exact enumeration when players (incl. shadows) ≤ 15; else sampled permutation with Monte-Carlo SE per player.
Diagnostics only	Benjamini-Hochberg p-values and Cohen's d are reported but DECIDE NOTHING (a permuted noise column can show BH p = 0.000).

A.6 Stage 3 — Selection (parsimony) · reviewed at gate G3

Eligible set	players importance_confirmed with confirmation frequency ≥ 0.80, ranked by Shapley value.
Parsimony curve	for k = 1..K fit on top-k players; record Gini/KS/AUC/pAUC/Brier/ECE with CIs on out-of-fold predictions (same rows → DeLong's paired test valid). Full curve is a required output.
Stopping rule (pre-declared)	smallest k that is BOTH (a) not distinguishable from the full-candidate model by DeLong's test at α = 0.05, AND (b) within ≤ 5% relative Gini loss. If none qualifies, that is the reported result.

A.7 Stage 4 — Calibration (always on) · reviewed in gate G4

Calibration	isotonic regression via cross-validated CalibratedClassifierCV (map on held-out predictions). Report Brier, ECE, reliability diagram.
Usage rule	calibrated probabilities used whenever a PD is quoted; raw scores valid for ranking only.

A.8 Deployment scoring & controls · prepared at gate G5

Exact additive score	log-odds = intercept + Σ (coef_j × WOE_j) → faithful per-feature reason codes, no post-hoc approximation.
PD floor & cap (always on)	pd_floor = 0.0003 (0.03%, Basel CRE32.3) · pd_cap = 0.999; applied at scoring only.
Omission floor	applicant-influenceable columns with protective missing-WOE floored to the most-adverse observed WOE (applied AFTER fit).
Mandatory gate (primary)	an application missing a declared-essential field is not scored → REFER / REJECT.
Decisions emitted	SCORE / SCORE_FLAGGED / REFER / REJECT, each with a calibrated PD + reason string.

A.9 Per-grade calibration backtest (inside AEGIS gate G4)

Split	80/20 stratified, random_state = 42.
Grades	PD → grade via rating_masterscale.pd_to_grade (institution internal master scale).
Per grade	observed default rate with Wilson 95% CI vs mean predicted PD (in-CI check).
Calibration omnibus	Hosmer-Lemeshow test ($p < 0.05$ rejects adequate calibration); ECE reported.
Fairness (consumer)	Adverse Impact Ratio, EEOC four-fifths (AIR ≥ 0.80) across protected-attribute bands; age under ECOA/Reg B EDDSS exception (no negative factor for applicants 62+).

A.10 AEGIS governance — gates, hash chain, materiality tiers

Gates	G0 scope/eligibility · G1 data sufficiency · G2 attribution · G3 selection · G4 performance/calibration/fairness · G5 deployment governance · GF final Go/No-Go. Each: machine recommendation + evidence + 2–4 options → one human signer.
Hash chain	entry_hash = SHA-256(prev_hash + decision_content)[:8]; genesis = GEN-<TAG>0 — a tamper-evident decision ledger.
EPV	events per variable = defaults ÷ features; sufficiency rule EPV ≥ 10 (Peduzzi 1996).

Materiality tiers	LOW (immaterial; owner batch-sign) · MEDIUM (benchmark; stage gates + MAC notice) · HIGH (core capital-driving; independent validator every gate; strict bars EPV≥10, AUC≥0.80, HL pass, no thin grades n≈20).

A.11 All fixed thresholds at a glance

Seed (all stages)	42
CV folds	5 (stratified, shuffled)
IV screen	≥ 0.02
Non-monotonic re-route	≥ 20% IV recovery
χ^2 reliability gate	min expected cell ≥ 5
Correlation cluster cut	|Spearman ρ| = 0.7 (sens. 0.6/0.7/0.8)
VIF flag	> 10
Shadow players	≥ 10
Shadow-null bar	95th percentile of shadow Shapley
Confirmation frequency	≥ 0.80
Exact Shapley cutoff	≤ 15 players (incl. shadows)
Parsimony DeLong	α = 0.05 (not distinguishable)
Parsimony Gini tolerance	≤ 5% relative loss
Calibration	isotonic (cross-validated)
PD floor / cap	0.03% (0.0003) / 0.999
EPV sufficiency	≥ 10
Fairness AIR	≥ 0.80 (four-fifths)
Backtest split	80/20 stratified, seed 42

A.12 Metric glossary (one line each)

AUC	area under ROC; P(model ranks a random bad above a random good). 0.5 = chance.
Gini	2·AUC − 1; rank-ordering power. 0 = chance, 1 = perfect.
KS	maximum gap between good/bad cumulative score distributions.
Brier	mean squared error of predicted probabilities (lower is better).
ECE	expected calibration error; mean |predicted − observed| across PD bins.
IV	Information Value; WOE-weighted univariate separation of good vs bad.
Cramér's V	strength of categorical association (0–1).
Shapley value	a feature's fair share of model performance over all feature coalitions.
DeLong test	significance test comparing two correlated AUCs.
Hosmer-Lemeshow	goodness-of-fit for calibration across PD bins.

Wilson CI	confidence interval for a proportion on small samples.
EPV	events (defaults) per variable; data-sufficiency measure.
AIR	adverse impact ratio; group selection-rate ratio (EEOC four-fifths ≥ 0.80).

A.13 Key references & regulatory sources

Williamson & Feng (2020), SPVIM (ICML) · Kursa & Rudnicki (2010), Boruta shadow features · DeLong, DeLong & Clarke-Pearson (1988), correlated-AUC test · Siddiqi (2006), IV bands / scorecards · Peduzzi et al. (1996), events-per-variable ≥ 10 · Hosmer & Lemeshow (1980) and Hosmer, Lemeshow & Sturdivant (2013), calibration GoF / AUC bands · Wilson (1927), score interval · Meinshausen & Bühlmann (2010), stability selection · Lessmann, Baesens, Seow & Thomas (2015), credit-scoring benchmark suite · Hand & Henley (1997) and Crook & Banasik (2004), accept-only samples / reject inference.

Regulatory: Basel CRE32.3 (PD floor 0.03%), CRE36.60 (use test), CRE36.79 (≥ 5yr PD history), CRE36.122–123 (calibration backtest) · EBA/GL/2017/16 §4.4 (Margin of Conservatism A/B/C) · SR 26-2 (Fed/OCC/FDIC, 17 Apr 2026; superseded SR 11-7) · ECOA / Regulation B §1002.6(b)(2) (EDDSS age exception) and EEOC four-fifths rule (AIR ≥ 0.80) · EU AI Act Annex III §5(b), high-risk credit scoring (deadline 2 Dec 2027).

Status caveats carried from the methodology: out-of-time validation and reject inference are DEFERRED (all datasets are i.i.d. public benchmarks); the cross-dataset benchmark and stability-selection studies are implemented but not yet run, so no comparative-superiority claim is made; grade-band edges and any published SOTA comparison require their own citation before publication.